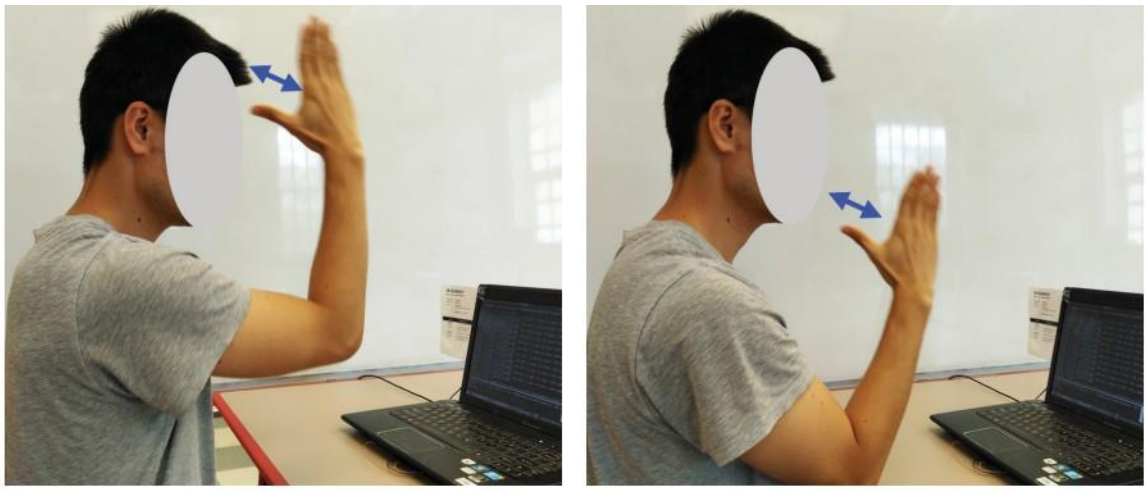
Sign Language Recognition Using WiFi Channel State Infiormation: Analysis ofi the SignFi Research Paper

1. **Introduction to Sign Language** Sign language is a complete visual-gestural language used by deaf and mute people. It employs hand shapes, arm movements, facial expressions, and body posture to convey meaning. Unlike spoken languages, sign languages are not universal; each country or region has its own distinct sign language with unique grammar, vocabulary, and syntax. Examples include American Sign Language (ASL), British Sign Language (BSL), and Indian Sign Language (ISL). In the United States, ASL serves as the primary language for approximately one-half to two million people and is recognized as a foreign language in many educational institutions. A sign language recognition system bridges the communication gap between deaf and hearing individuals by automatically translating gestures into text or speech.

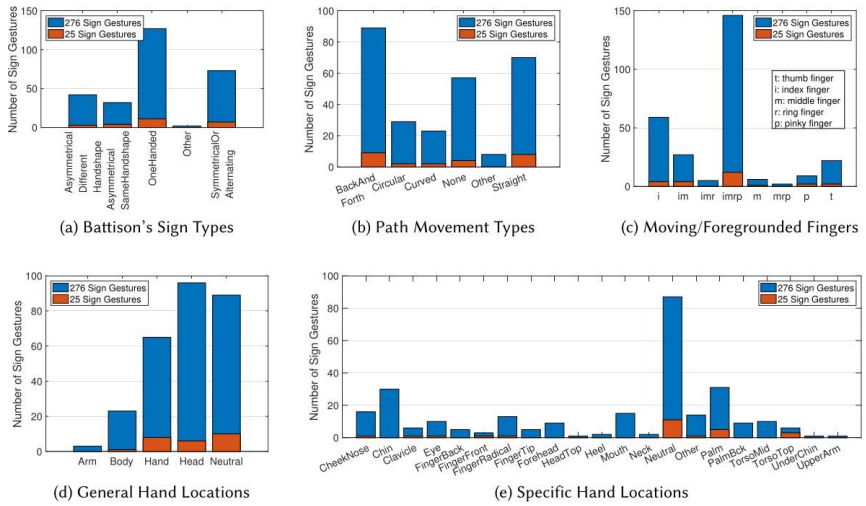


(a) Sign gesture for "Father" (b) Sign gesture for "Mother"

Fig. 2. Sign words "Father" and "Mother" have the same hand gesture and finger posture, but they need the dominate hand in different locations.

2. **Overview of the SignFi Research Paper** The paper titled "SignFi: Sign Language Recognition Using WiFi," published in Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (March 2018), presents SignFi, a non-intrusive, device-free system

that recognizes **American Sign Language (ASL)** gestures exclusively through commodity **WiFi** signals. SignFi addresses the limitations of existing technologies such as camera-based systems (sensitive to lighting), Kinect or Leap Motion sensors (limited to finger gestures and requiring specific positioning), and wearable gloves (intrusive). It is the first WiFi-based approach capable of recognizing **276 distinct ASL** gestures involving complex combinations of head, arm, hand, and finger movements.



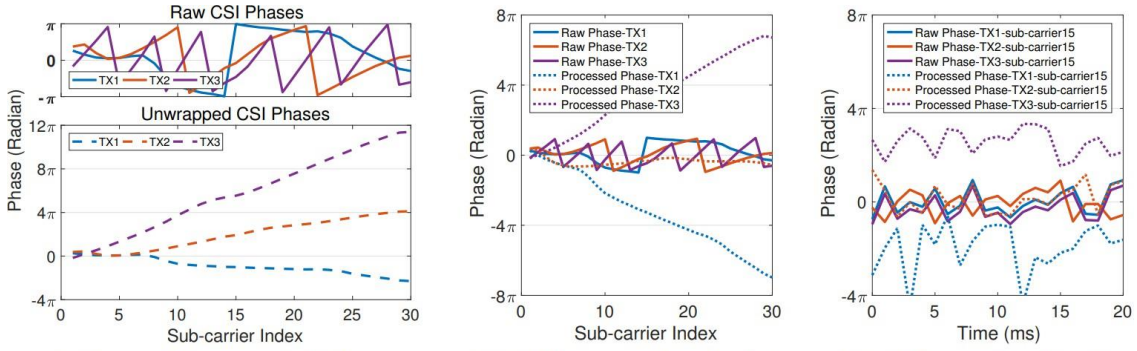
(a) Battison's Sign Types (b) Path Movement Types (c) Moving/Foregrounded Fingers

(d) General Hand Locations (e) Specific Hand Locations

3. **WiFi Setup and Technical Approach** SignFi operates using standard 802.11n WiFi hardware at the **5 GHz band** with a 20 MHz channel width.
The proposed setup consists of two laptops: one functions as the **WiFi Access Point** (AP) equipped with three external antennas, and the other as the **Station** (STA) with one internal antenna. The AP continuously transmits packets to the STA, and the STA extracts Channel State Information (CSI) from the packet preambles using a modified open-source 802.11n CSI tool.

CSI captures how wireless signals propagate through multiple paths and are reflected or scattered by human body movements. The system records CSI for 3 transmit antennas × 30 sub-carriers, producing amplitude and phase values. Raw CSI measurements undergo dedicated signal processing:

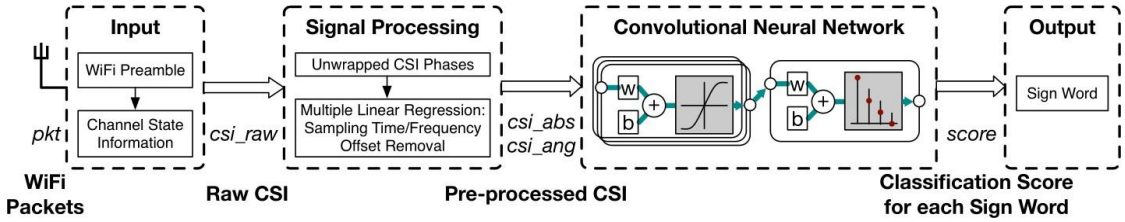
- Multiple linear regression removes random phase offsets caused by Sampling Time Offset (STO) and Sampling Frequency Offset (SFO).
- Phase unwrapping restores accurate information about CSI changes across sub-carriers and sampling time.



(a) Raw and Unwrapped CSI Phases (b) CSI Phase vs. sub-carrier Index (c) CSI Phase vs. Sampling Time

Fig. 6. Raw CSI measurements do not capture how CSI phases change over sub-carriers and sampling time.

The cleaned CSI traces (dimension: 200 time samples × 60 features (amplitude + phase) × 3 antennas) are fed into a 9-layer Convolutional Neural Network (CNN). The CNN architecture includes convolutional layers (3×3 kernels), batch normalization, ReLU activation, average pooling, dropout (probability 0.6), a fully connected layer, and softmax classification. Training uses Stochastic Gradient Descent with Momentum. This end-to-end learning approach automatically extracts spatio-temporal features unique to each sign gesture, eliminating the need for hand-crafted features required by earlier kNN + Dynamic Time Warping methods.



4. **Data Collection and Experimental Results** The researchers collected CSI traces for 276 frequently used basic ASL gestures selected from a standard ASL beginner vocabulary. Data was gathered from five male participants in both laboratory (13 m × 12 m) and home (4.11 m × 3.86 m) environments.

- **Lab environment**: 5,520 gesture instances (User 5: 20 repetitions per gesture; Users 1–4: 10 repetitions each for 150 gestures).
- **Home environment**: 2,760 gesture instances (10 repetitions per gesture).
- **Total dataset**: 8,280 labeled instances for 276 gestures.

Each participant watched instructional videos before performing gestures to ensure consistency. CSI was sampled approximately every 5 ms.

Performance results (5-fiold cross-validation on User 5): - Lab environment: 98.01 % accuracy

- Home environment: 98.91 % accuracy
- Combined lab + home: 94.81 % accuracy

User-independence test (7,500 instances ofi 150 gestures firom 5 users): 86.66 % average accuracy. These results demonstrate that SignFi significantly outperforms prior WiFi-based methods, which achieved only 68 % accuracy on the same 276 gestures using kNN + DTW.

Table 1. Number of Sign Words in Different Categories used in the Experiments

Common	Animals	Colors	Descriptions	Family	Food	Home	People	Questions	School	Time	Others	Total
16	15	12	32	31	54	17	13	6	26	31	23	276

Table 2. Data Collection Summary

	User	Age	Weight/Height	Data Collection Date (Number of Signs × Number of Repetitions)	Gesture Duration	Number of Instances
Lab	User 1	39	90kg/170cm	Oct. 18, 2017 (25×10); Nov. 2, 2017 (125×10)	1s-2.5s	1,500
	User 2	28	61kg/174cm	Oct. 18, 2017 (25×10); Oct. 30, 2017 (125×10)	0.5s-1.5s	1,500
	User 3	31	55kg/168cm	Oct. 21, 2017 (25×10); Nov. 6, 2017 (125×10)	0.5s-1.5s	1,500
	User 4	26	65kg/180cm	Oct. 23, 2017 (25×10); Oct. 31, 2017 (125×10)	1s-2.5s	1,500
	User 5[a]	29	68kg/171cm	Jul. 18, 2017 (166×20); Jul. 19, 2017 (110×20)	0.5s-1.5s	5,520
Home	User 5	29	68kg/171cm	Jun. 8, 2017 (32×10); Jun. 25, 2017 (68×10); Jul. 4, 2017 (100×10); Jul. 11, 2017 (25×10); Jul. 12, 2017 (51×10)	0.5s-1.5s	2,760

5. **Limitations Regarding Idle and Activity Detection** SignFi does not address the detection of idle states or differentiation between signing and other human activities. All evaluations were performed on manually segmented CSI traces, where gesture start and end points were explicitly marked. The system assumes the input is a valid sign gesture and contains no mechanism for continuous streaming data that includes periods of inactivity or unrelated movements. The paper explicitly notes that automatic time segmentation and idle/activity detection are left for future work.

6. **Proposed Ideas fior Idle and Activity Detection** To enable practical real-time deployment, the following lightweight extensions can be integrated with the existing SignFi pipeline:

a. **Variance-based Threshold Detector** Continuously monitor the short-term variance or total energy of CSI amplitude across sub-carriers. When variance exceeds a predefined adaptive threshold for a minimum duration (e.g., 0.5–1 second), mark the segment as potential activity and forward it to the CNN classifier. When variance drops below the threshold, classify the

period as idle.

b. **Extended CNN with Idle Class** Augment the training dataset with idle-period CSI traces (person sitting still) and expand the output layer to 277 classes (276 ASL signs + 1 "idle" class). The softmax layer will then directly output whether the person is idle or performing a specific sign.

c. **Two-Stage Architecture** Stage 1: A low-complexity activity detector (threshold or lightweight neural network) identifies movement segments in real time.

Stage 2: Only detected segments are passed to the full 9-layer SignFi CNN for sign classification. This approach minimizes computational load during idle periods.

These enhancements preserve the core SignFi accuracy while enabling robust, continuous operation in unconstrained environments.

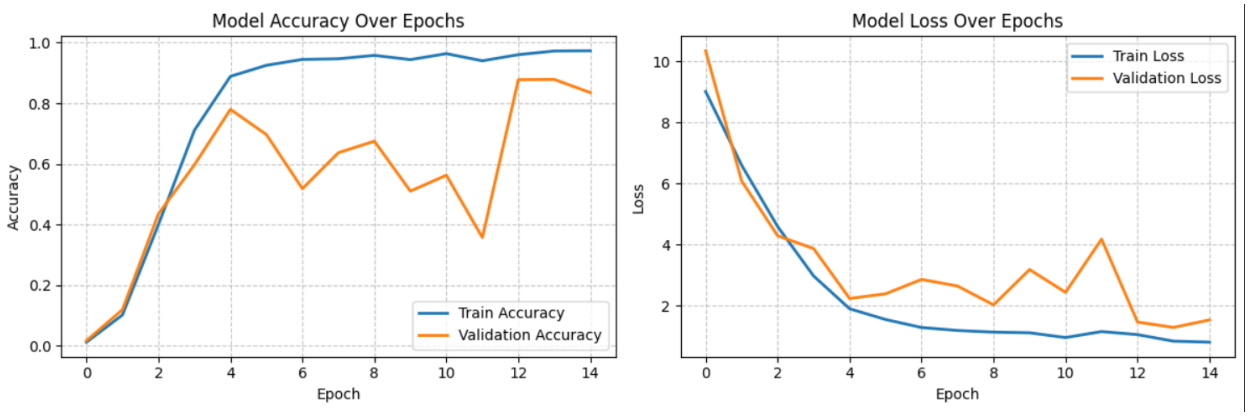
7. Our CNN Implementation & Experimental Results (Colab Link)

We reproduced the SignFi pipeline using TensorFlow/Keras on the lab dataset, covering all 276 ASL gestures across the full set of recorded instances. The raw CSI data was transposed and processed to extract both amplitude and phase components, which were concatenated to form a six-channel input tensor spanning 200 time steps and 30 sub-carriers. An 80/20 stratified train-test split was applied to ensure balanced class distribution across training and evaluation sets.

Model Architecture

Our CNN consists of a single convolutional block followed by a dense classifier: a Conv2D layer (4 filters, 4×4 kernel), BatchNormalization, ReLU activation, MaxPooling2D (4×4, stride 4×4), Flatten, and a Dense output layer (276 classes, softmax) with L2 regularization ($\lambda = 0.01$). Total trainable parameters: 325,248 (1.24 MB). The model was compiled with SGD (learning rate = 0.01, momentum = 0.9) and sparse categorical crossentropy loss, trained for 10 epochs with batch size 32.

Training Results



The model converged quickly, reaching a peak validation accuracy of 91.39% at epoch 6. By epoch 10, training accuracy was 92.32% and validation accuracy stabilized at 88.41%. Training loss dropped from 8.91 at epoch 1 to 1.34 by epoch 10, with validation loss settling at 1.55.

Final Test Accuracy: 88.41%

Comparison with the Original Paper

The original SignFi paper reports 98.01% accuracy (lab environment, 5-fold cross-validation on User 5) using a deeper 9-layer CNN with average pooling and dropout ($p = 0.6$). Our simplified single-layer model achieves 88.41% — a difference of ~10%, which is expected given the shallower architecture and single train-test split used in our implementation. The results nonetheless confirm that WiFi CSI amplitude and phase features are sufficient for high-accuracy sign language recognition even with a lightweight model.